HACKATHON LOGARITHM

Quantum Water Allocation Under Drought Scenarios

A QAOA approach to **SDG 6.4** — sustainable water allocation in the Alto Atoyac Basin, Puebla, México.

TEAM 11

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REPOSITORY `github.com/ludago4499/Hackathon-LATAM-OQI`

The problem — and why it matters (SDG 6.4)

The decision. Under drought, allocate scarce water among competing users while protecting sources and **prioritizing human consumption**.

SDG 6.4 — "substantially increase water-use efficiency and ensure sustainable withdrawals to address water scarcity." We optimize *who is cut* when supply cannot meet everyone.

Benchmark objective — minimize the NWWD

$$\mathrm{NWWD} = w_u \sum_j \frac{u_j^{\text{urb}}}{D_j^{\text{urb}}} + w_a \sum_j \frac{u_j^{\text{agri}}}{D_j^{\text{agri}}}$$

u_j = unmet demand · D_j = demand · weights w_u=10 (urban), w_a=3 (agri).

Key fact about the instance

Total demand **149** hm^3/yr, but max supply is only Normal 125 Moderate 100 Severe 75.

Supply never meets demand — the optimizer must choose who goes without.

Sources. Valle de Puebla aquifer 80, Atlixco–Izúcar 45 (hm^3/yr).

Drought factor α: $A_i^{\text{eff}} = \alpha A_i$, with $\alpha = 1.0/0.8/0.6$.

Classical baseline — exact MILP / LP (our ground truth)

Formulation (benchmark uses $\lambda = 0 \Rightarrow$ pure LP):

$$\min \mathrm{NWWD} + \lambda \sum_{i,j} c_{ij}\, x_{ij}$$

$$\begin{aligned}
\text{s.t.}\quad & \textstyle\sum_j x_{ij} \le A_i^{\text{eff}} = \alpha A_i && \text{(capacity)}\\
& \textstyle\sum_i x_{ij} + u_j = D_j && \text{(demand balance)}\\
& x_{ij} \ge 0,\ u_j \ge 0 && \text{(solved exactly, CBC)}
\end{aligned}$$

Benchmark instance (Annex A):

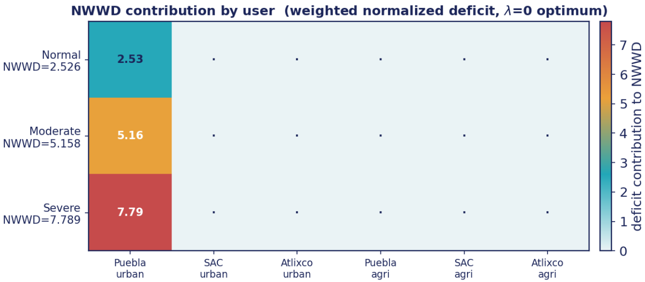
Municipality	Urban D	Agri D
Puebla	95	8
San Andrés Cholula	14	5
Atlixco	9	18

Validated optimum ($\lambda = 0$, exact LP)

Scenario	NWWD	**unmet**
NORMAL	**2.526**	24
MODERATE	**5.158**	49
SEVERE	**7.789**	74

These three numbers are the **target** the quantum solver is measured against. All unmet volume lands on **Puebla urban** — next slide.

Where the deficit lands — reading the NWWD



The color map decomposes the NWWD into each user's weighted, normalized deficit at the optimum.

Every scenario's entire deficit sits on Puebla urban. Because the metric normalizes by demand, Puebla's urban penalty per hm^3 is $10/95 \approx 0.11$ — the *cheapest* to leave unmet.

This is an honest property of the benchmark, and the seed of our critique (later): the objective is efficient, but not automatically **fair**.

From allocations to qubits — discretization & QUBO

1 · Discretize. Each continuous allocation → blocks of B hm³, binary-encoded:

$$x_{ij} = B \sum_k 2^k \, b_{ijk}, \qquad b_{ijk} \in \{0, 1\}$$

2 · Slack = unmet. Demand balance becomes a *quadratic penalty*; the slack *is* u_j, so the objective is **literally the** NWWD.

3 · Standard form.

$$\min_{b \in \{0,1\}^n} \; b^\top Q \, b \;=\; \underbrace{\mathrm{NWWD}}_{\text{objective}} + \underbrace{\text{constraints}}_{\text{penalties}}$$

Why it maps naturally to a QUBO

- binary allocation decisions after discretization
- combinatorial search space that grows fast
- source → demand *graph* structure
- constraints = quadratic penalties

B is the key knob (real counts)

B (hm³)	20	10	5
qubits (full)	24–28	32–36	49–53
qubits (urban)	15–19	20–24	31–35

smaller B = finer resolution, more qubits.

Penalties $\to$ Ising Hamiltonian $\to$ QAOA ansatz

Constraints $\to$ quadratic penalties

$$P\Big(\sum_j x_{ij} - A_i + \text{slack}_i\Big)^2 \qquad \text{(capacity)}$$

$$P\Big(\sum_i x_{ij} - D_j + u_j\Big)^2 \qquad \text{(demand)}$$

Penalty P scaled per block so feasibility dominates the deficit *without* burying the NWWD signal.

QUBO $\to$ **Ising** via $x_i \mapsto (1 - Z_i)/2$:

$$H_C = \sum_i h_i Z_i + \sum_{i<j} J_{ij} Z_i Z_j + \text{const}$$

Minimizing NWWD $\Leftrightarrow$ **ground state of H_C**.

QAOA ansatz

Initial state: equal superposition $|+\rangle^{\otimes n}$.

Cost: $U_C(\gamma) = e^{-i\gamma H_C}$ — phase by NWWD.

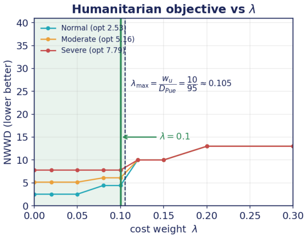
Mixer: $U_M(\beta) = e^{-i\beta H_M}$, $H_M = \sum_i X_i$.

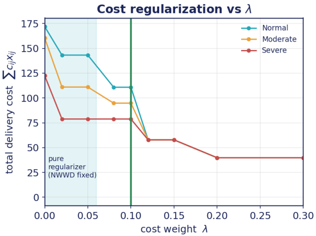
$$|\psi(\boldsymbol{\gamma}, \boldsymbol{\beta})\rangle = \prod_{l=1}^{p} U_M(\beta_l)\, U_C(\gamma_l)\ |+\rangle^{\otimes n}$$

Classical optimizer (COBYLA) tunes $(\boldsymbol{\gamma}, \boldsymbol{\beta})$ over $2p$ angles; sample the best bitstring.

PennyLane statevector; backend swap (sim $\to$ GPU $\to$ QPU) isolated in `backend.py`.

Choosing $\lambda = 0.1$ — a principled operating point





The cost term $\lambda \sum c_{ij} x_{ij}$ needs the right weight. We derive it, not guess it.

Dominance bound. Human priority is preserved iff $\lambda < (w_d/D_j)/c_{ij}$ on every served edge. The binding edge is **Puebla-urban via Valle de Puebla**:

$$\lambda_{\max} = \frac{w_u}{D_{\text{Pue}}} = \frac{10}{95} \approx 0.105.$$

Verified by MILP sweep. Severe keeps the *exact* optimum up to $\lambda^* = 0.106$.

Why 0.1? Largest round value below the ceiling: keeps human demand first in the worst-case drought, **maximizes cost-regularization**, and lifts the $\lambda=0$ degeneracy $\Rightarrow$ a unique min-cost allocation and a better-conditioned QAOA landscape.

Sizing each instance — qubit budget known up front

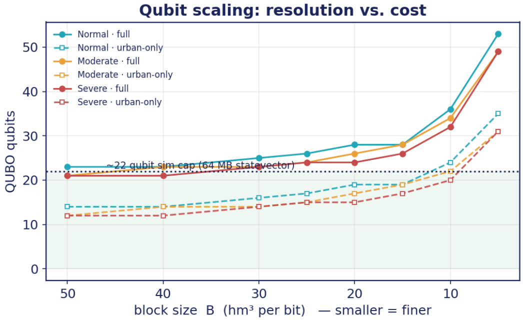
Per decision variable, a range $[a, b]$ discretized at block size B costs

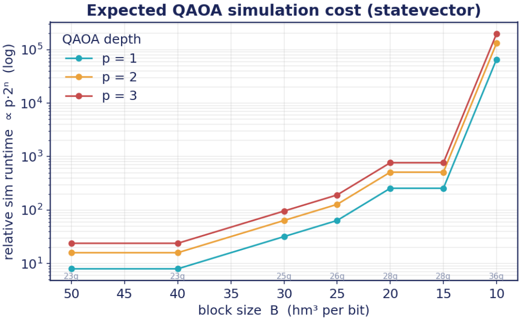
$$n_{\text{bits}} = \left\lceil \log_2 \left(b - a + 1 \right) \right\rceil, \qquad b - a = \left\lceil \frac{\text{range}}{B} \right\rceil \text{ blocks}, \qquad n_{\text{q}} = \sum_{\text{vars}} n_{\text{bits}} + \text{slack}.$$

B (hm³)	Normal	Moderate	Severe
25	23	23	21
20	28	24	24
18	28	26	24
15	28	28	24
13	28	28	26
12	28	28	28
10	33	29	29
1	93	93	89

Same code sweeps many benchmarks by varying **scenario, B, p, max_iter, penalty, seed, lambda, restarts, urban_only**. Computing n_{q} **beforehand** tells us whether a test is realistically runnable — the statevector needs $2^n \cdot 16$ B of RAM (~22-qubit cap here).

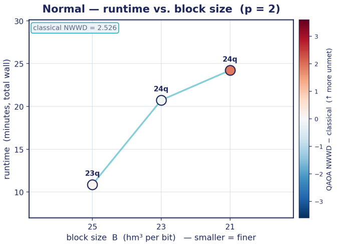
Benchmarking I — scaling: resolution vs. cost

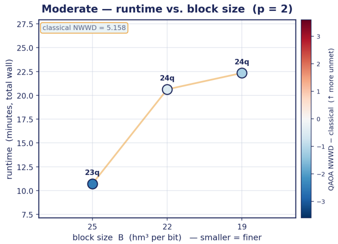


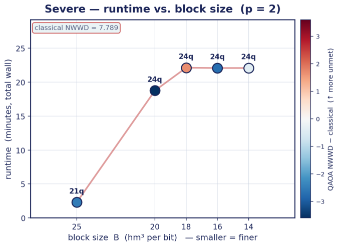


Left: real QUBO qubit counts. The *urban-only* + slack-free capacity variant keeps small B under the **~22-qubit** simulator ceiling. **Right:** exact statevector cost grows as $p \cdot 2^n$ — the scaling wall that motivates the lean model and the two-stage method (backup). **Measured (B, p) runtimes drop onto this curve.**

Benchmarking II — runtime vs. block size B







What the runs show

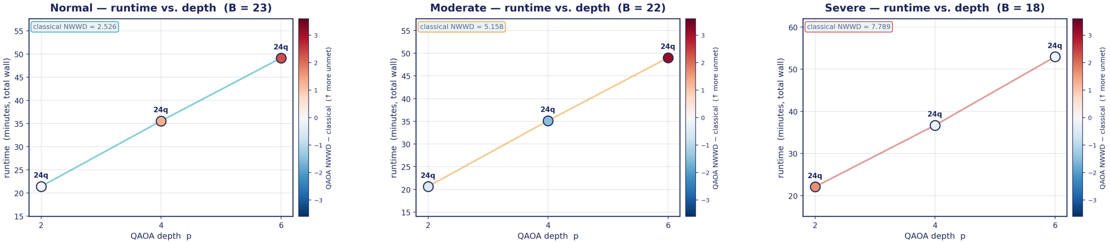
- runtime tracks the **qubit count**, not B: 24q $\approx$ 20–22 min, dropping to 2–11 min once B frees a qubit (21–23q)
- point colour $=$ $\mathrm{QAOA\ NWWD}$ vs the **exact optimum**; label $=$ qubits
- feasibility is enforced by the penalty P scaling

Honest framing

The goal is **not** quantum advantage — classical MILP wins at this scale. We test whether the *structure* suits future quantum optimization: native QUBO/Ising fit, tunable depth p, same metric $\Rightarrow$ a fair benchmark.

Marker colour $=$ $\mathrm{QAOA\ NWWD}$ $-$ classical optimum (blue below, red $=$ more unmet); qubit count labelled. Total wall time, PennyLane statevector.

Benchmarking II — depth sweep (runtime vs. p)



At fixed B, deeper circuits cost **roughly linearly** more wall time ($p\!:\!2\to 6$) yet **do not improve** NWWD— across all three scenarios it drifts *further* from the optimum. The depth/accuracy trade-off is not free.

Same encoding as previous slide: marker colour = QAOA NWWD − classical optimum; qubit count labelled.

Critiques, fixes, and how we used AI

Critiques of the problem statement

- **Fairness**: normalization dumps all shortage on the biggest city (Puebla urban).
- **Single objective**: folding human + economic goals into one weighted sum hides a Pareto trade-off.
- **Discretization** truncates continuous volumes to a few bits.
- **Noise**: real hardware out of scope — we simulate.

Our fixes

- add a min-max / per-capita deficit term for fairness
- λ sweep $\rightarrow$ report the humanitarian$\leftrightarrow$cost curve (done)
- finer / non-uniform B where truncation hurts

Advantages

clean QUBO/Ising fit · same NWWD $\Rightarrow$ honest benchmark · depth $p =$ tunable scaling study · penalties encode all constraints.

Disadvantages

no advantage at this size · discretization loss · qubits grow as $S \cdot D \cdot B$+slack · noise-sensitive (simulated only).

How we used AI

speed up QUBO-mapping & explanation · organize benchmark tables · draft/structure slides. **Every number verified against the LP baseline.**

Takeaway

We built an end-to-end, honest benchmark.

- **Formulation** — NWWD tied to SDG 6.4; supply never meets demand.
- **Baseline** — exact LP: **2.526 / 5.158 / 7.789**.
- **Quantum** — QUBO $\rightarrow$ Ising $\rightarrow$ QAOA, backend-agnostic.
- $\lambda = 0.1$ — derived from the dominance bound $10/95$.
- **Benchmark** — same metric, scaling + direct comparison.

Repository

`github.com/ludago4499/Hackathon-LATAM-OQI`

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Gracias · Thank you

Backup — two-stage hierarchical QAOA

Idea. Run a **coarse** QAOA at $B_0{=}20$; bracket each variable to $[\,v - \text{margin}\cdot B_0,\ v + \text{margin}\cdot B_0\,]$; re-encode over the narrow window (offset encoding $x = \text{lo} + B_1 \sum_k 2^k b_k$) and run a **fine** QAOA at B_1. Fewer qubits at finer resolution.

Two questions, separated:

(A) *ceiling* — does the bracket still contain the optimum? (brute-force)

(B) *solver* — does QAOA find it? (sampled NWWD)

Findings (urban-only)

- margin $= 1$: bracket **valid in every checked case** (contains optimum).
- best case **Severe** $B_1{=}5$: full $= 24$q, bracketed $=$ **9**q, same 6.84 optimum.
- margin $= 0$ over-prunes; coarse-stage error dominates $\Rightarrow$ spend budget there / seed from LP.

```
python -m qaoa.hierarchical --scenario Severe --B0
20 --B1 10 --margin 1
```